Hypothesis Testing

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Basics of Hypothesis Testing Concepts

1. **Null Hypothesis (H_0)** No effect or no difference. *In ML we say baseline model or status quo (e.g., "new model is not better")*
2. **Alternative Hypothesis (H_1)** There is an effect or difference. *i.e., claim that new model/feature improves performance*
3. **Test Statistic** A numerical value (like z, t) used to test hypotheses. *i.e., performance metric difference, e.g., accuracy gap*
4. **P-value** Probability of observing data as extreme as sample under H_0. *i.e., how surprising the result is assuming no real difference*
5. **Significance Level (α)** Threshold for rejecting H_0 (commonly 0.05). *i.e., confidence required to claim improvement is real*

Basics of Hypothesis Testing Concepts Cont'd...

6. **Type I Error (α)** Rejecting H_0 when it's true (false positive). *i.e., believing model is better when it's not*
7. **Type II Error (β)** Failing to reject H_0 when H_1 is true (false negative). *i.e., missing a real improvement*
8. **Power of a Test ($1 - \beta$)** Probability of correctly rejecting false H_0. *i.e., sensitivity — detecting real improvement*
9. **One-tailed Test** Tests for effect in one direction. *i.e., is the new model better?*
10. **Two-tailed Test** Tests for effect in both directions. *i.e., is the new model either better or worse?*

Hypothesis Testing using Z-Test (Examples)

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Right-Tailed Z-Test (ML Accuracy)

Question

A new spam classifier claims to have an accuracy higher than the existing model (85%). From 50 test runs, the mean accuracy was 88% with a known population std. dev. of 5%. Test the claim at a 5% level of significance.

Step 1: Hypothesis Formulation $H_0 : \mu = 85\%$, $H_1 : \mu > 85\%$ (right-tailed test)
Step 2: Level of Significance $\alpha = 0.05$, Critical $z = z_{0.05} = 1.645$
Step 3: Test Statistic

$$z = \frac{\bar{X} - \mu_0}{SE(\bar{X})} = \frac{88 - 85}{5/\sqrt{50}} = \frac{3}{0.707} \approx 4.24$$

$$p\text{-value} = P(Z > 4.24) \approx 0.00001$$

Conclusion: $z = 4.24 > 1.645 \Rightarrow$ Reject H_0. *Classifier has significantly better accuracy.*

Left-Tailed Z-Test (Precision Drop)

Question

You suspect a model's precision dropped from the expected 90%. From 36 runs, average precision is 87%. Population std. dev. is 6%. Test this claim at the 5% level of significance.

Step 1: $H_0 : \mu = 90\%$, $H_1 : \mu < 90\%$ (left-tailed test)
Step 2: $\alpha = 0.05$, Critical $z = -1.645$

$$z = \frac{87 - 90}{6/\sqrt{36}} = \frac{-3}{1} = -3$$

$$p\text{-value} = P(Z < -3) \approx 0.00135$$

Conclusion: $z = -3 < -1.645 \Rightarrow$ Reject H_0. *Model precision has significantly dropped.*

Two-Tailed Z-Test (F1-Score Shift)

Question

You evaluate if a new regularization technique changes the F1-score from baseline 0.75. From 25 runs, the mean score is 0.78. Known population std. dev. is 0.06. Use $\alpha = 0.05$ to test.

Step 1: $H_0 : \mu = 0.75$, $H_1 : \mu \neq 0.75$ (two-tailed test)
Step 2: $\alpha = 0.05 \Rightarrow z_{0.025} = \pm 1.96$

$$z = \frac{0.78 - 0.75}{0.06/\sqrt{25}} = \frac{0.03}{0.012} = 2.5$$

$$p\text{-value} = 2 \cdot P(Z > 2.5) \approx 2 \cdot 0.0062 = 0.0124$$

Conclusion: $|z| = 2.5 > 1.96 \Rightarrow$ Reject H_0. *F1-score has significantly changed.*

Hypothesis Testing for ML Model Accuracy

Question

A machine learning model for image classification was trained with a known population accuracy of 85%. A new batch of 36 test samples gives a sample mean accuracy of 83%, with known population standard deviation $\sigma = 4\%$. Test at the 5% significance level if the new batch has lower accuracy.

Known:

- $\mu_0 = 85\%$
- $\bar{x} = 83\%$
- $\sigma = 4\%$, $n = 36$
- $\alpha = 0.05$

Use: One-sample z-test (right-tailed)

Solution:

Step 1: State Hypotheses

- Null Hypothesis: $H_0 : \mu = 85\%$
- Alternative Hypothesis: $H_1 : \mu < 85\%$ (Left-tailed test)

Step 2: Compute Test Statistic

$$
\begin{aligned}
z &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \\
&= \frac{83 - 85}{4/\sqrt{36}} = \frac{-2}{4/6} = \frac{-2}{0.6667} \approx -3.00
\end{aligned}
$$

Step 3: Rejection Region

- Significance level: $\alpha = 0.05$
- Critical value for left-tailed test: $z_{0.05} = -1.645$
- Rejection region: $z < -1.645$
- **Observed** $z = -3.00 \in$ Rejection Region

Solution Cont'd...

Step 4: Compute p-value

$$\begin{aligned} \text{p-value} &= P(Z < -3.00) \\ &= 0.0013 \quad \text{(from standard normal table)} \end{aligned}$$

Since p-value $< \alpha$, we reject H_0.
There is sufficient evidence at the 5% significance level to conclude that the new batch has **significantly lower** accuracy than the original model.

Compute β

Assume true mean under H_1 is $\mu = 82\%$

Critical value:

$$\bar{x}_c = \mu_0 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}} = 85 + (-1.645) \cdot \frac{4}{6} = 83.903$$

Beta is the probability of failing to reject H_0 when $\mu = 82\%$:

$$\beta = P(\bar{x} > 83.903 \mid \mu = 82) = P\left(Z > \frac{83.903 - 82}{4/\sqrt{36}}\right)$$

$$= P(Z > 4.76) \approx 0.000001$$

Hypothesis Testing for Two Populations (Examples)

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Example 1: $\mu_1 > \mu_2$

Question:

A company trains two image classifiers:

- Model A: trained with data augmentation.
- Model B: trained without data augmentation.

Over 50 trials:

$$\bar{x}_1 = 87\%, \quad \sigma_1 = 3\%, \quad n_1 = 50$$

$$\bar{x}_2 = 85\%, \quad \sigma_2 = 4\%, \quad n_2 = 50$$

At a 5% level of significance, test whether the augmented model (A) performs significantly better than the non-augmented model (B).

Step 1: Hypotheses Formulation

$H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 > \mu_2$

Z-test and p-value

$$z = \frac{87 - 85}{\sqrt{(3^2/50) + (4^2/50)}} = \frac{2}{\sqrt{0.18 + 0.32}} = \frac{2}{\sqrt{0.5}} \approx 2.83$$

$$\text{Critical value: } z_{0.05} = 1.645 \quad \Rightarrow \quad \textbf{Reject } H_0$$

$$\text{p-value} = P(Z > 2.83) \approx 1 - 0.9977 = 0.0023$$

Conclusion: p-value $< \alpha \rightarrow$ Model A is significantly better.

Example 2: $\mu_1 < \mu_2$

Question:

A researcher tests if removing a noisy feature from an ML model improves performance.

- Model A (with feature): $\bar{x}_1 = 84\%$, $\sigma_1 = 4\%$, $n_1 = 40$
- Model B (without feature): $\bar{x}_2 = 86\%$, $\sigma_2 = 3\%$, $n_2 = 40$

At $\alpha = 0.01$, test whether Model B has significantly higher accuracy than Model A.

Hypotheses: $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$, $\alpha = 0.01$

Z-test and p-value

$$z = \frac{84 - 86}{\sqrt{(4^2/40) + (3^2/40)}} = \frac{-2}{\sqrt{0.625}} \approx -2.53$$

$$\text{Critical value: } z_{0.01} = -2.33 \quad \Rightarrow \quad \textbf{Reject } H_0$$

$$\text{p-value} = P(Z < -2.53) \approx 0.0057$$

Conclusion: p-value $< \alpha \rightarrow$ Removing the feature improves the model.

Example 3: $\mu_1 \neq \mu_2$

Question:

An ML engineer evaluates the model's robustness across two domains:

- Domain A: $\bar{x}_1 = 90\%$, $\sigma_1 = 3.5\%$, $n_1 = 60$
- Domain B: $\bar{x}_2 = 88.5\%$, $\sigma_2 = 4\%$, $n_2 = 60$

At $\alpha = 0.05$, test if the accuracy differs significantly between domains.

Step 1:Hypotheses Formulation

$H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$, $\alpha = 0.05$

Z-test and p-value

Step 2: Level of Significance

$$\text{Critical values: } \pm 1.96 \quad \Rightarrow \quad \textbf{Reject } H_0$$

$$z = \frac{90 - 88.5}{\sqrt{(3.5^2/60) + (4^2/60)}} = \frac{1.5}{\sqrt{0.2042 + 0.2667}} = \frac{1.5}{0.686} \approx 2.19$$

$$\text{p-value} = 2 \cdot P(Z > 2.19) = 2 \cdot (1 - 0.9857) = 0.0286$$

Conclusion: p-value $< \alpha \rightarrow$ Significant difference in accuracy across domains.